TEMPLATE WEB FINAL — CSE 4165

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1. JAVASCRIPT

```
// Basic Output & Reading Inputs:
let output = "";
output += "Average = " + result + "<br>";
document.getElementById("output").innerHTML = output;
let name = document.getElementById("name").value; // [reading from an input]
let marks = Number(document.getElementById("marks").value);
```

```
// I have an array. Some STLs (Array Methods):
.length, .push("") [adds to end], .includes(""), .sort(), .join(", "),
.pop() [removes from end], .unshift("") [adds to the beginning],
.shift() [removes from the beginning], .indexOf(""), .reverse()
```

```
// JavaScript Objects:
let student = {
    name: "Swaron",
    id: "0112230418",
    cgpa: 3.63
};

// Properties can be accessed in 2 ways:
let sName = student.name;
// or
let sId = student["id"];

// JSON Conversion:
// 1. Convert JS Object -> JSON String
let jsonString = JSON.stringify(student);

// 2. Convert JSON String -> Back to JS Object
let parsedStudent = JSON.parse(jsonString);
// Now you can access parsedStudent.name or parsedStudent.cgpa!
```

2. HTML & DOM

```
<!-- Text input (for Subject Name, Password, etc.) -->
<input type="text" id="subjectInput" placeholder="Enter Subject">

<!-- Number input (for Study Hours, Calories, Heart Rate, etc.) -->
<input type="number" id="hoursInput" placeholder="Enter Hours">

<!-- Button & Output Div & Script Connection -->
<button onclick="startExam()">Submit</button>
<div id="output"></div>
<script src="script.js"></script>
```

```
// IMPORTANT RULE:
// For normal tags like <div>, <p>, <label>, <h3> (which have an opening and closing tag), you use .innerHTML or .innerText.
// For <input> boxes, you NEVER use .innerHTML. You use .value!
```

```
// Reading text input (String) -> just use .value
let subject = document.getElementById("subjectInput").value;

// Reading number input (Int or Float) -> wrap in Number()!
let hours = Number(document.getElementById("hoursInput").value);

// Reading text from a normal div:
<div id="myDiv">Hello World</div>
const value = document.getElementById("myDiv").textContent;
```

```
// JS Math Helpers:
// In JS, bmi.toFixed(2) rounds any number to 2 decimal places.
// Absolute Value: Math.abs(x)   | Rounding Down: Math.floor(x) | Rounding Up: Math.ceil(x)
// Math.random() gives a decimal between 0 and 0.999...
// Random integer between min and max: Math.floor(Math.random() * (max - min + 1)) + min;
```

```
<!-- 1. Dropdown (<select>) -->
<select id="location">
    <option value="">Select Location</option>
    <option value="Dhaka">Dhaka</option>
    <option value="Chittagong">Chittagong</option>
</select>

<!-- 2. Radio Buttons (Give both the SAME name="area" so user can't pick both!) -->
<input type="radio" name="area" id="urban" value="Urban">
Urban
<input type="radio" name="area" id="rural" value="Rural">
Rural

// Reading Dropdown, Radio Buttons & Disabling Input:
let loc = document.getElementById("location").value;
let isUrban = document.getElementById("urban").checked;
let isRural = document.getElementById("rural").checked;
document.getElementById("guessInput").disabled = true;
```

3. PHP

```
// Browser URLs:
http://localhost/web_final/rover.php   |
http://localhost/phpmyadmin/
```

```php
<?php
$attendees = 150;
$capacity = 60;
$name = "Swaron";
?>

<!-- HTML Form for PHP -->
<form action="process.php" method="POST">
    <input type="number" name="students" placeholder="Number
of Students">
    <input type="number" name="slicesPerStudent"
placeholder="Slices per Student">
    <input type="number" name="slicesPerPizza"
placeholder="Slices per Pizza">
    <button type="submit">Calculate</button>
</form>

<?php
// Grab the 3 inputs from the HTML form using their 'name'
attribute
$students = $_POST["students"];
$slicesPerStudent = $_POST["slicesPerStudent"];
$slicesPerPizza = $_POST["slicesPerPizza"];

// Useful PHP Math Functions:
// min($a, $b, $c)  -> finds minimum
// max($a, $b, $c)  -> finds maximum
// ceil($a / $b)    -> rounds UP to next whole integer
// (int)($a / $b)   -> integer division (for digit extraction
loop)

// PHP Array & Loop:
$fruits = ["Apple", "Banana", "Mango", "Orange"];

for ($i = 0; $i < count($fruits); $i++) {
    echo $fruits[$i] . "<br>";
}
?>
```

4. SQL & PHP MYSQLI

```sql
-- 1. Create and select the database
CREATE DATABASE uiu_bookshop;
USE uiu_bookshop;

-- 2. Create the table (INT for whole numbers, VARCHAR(100)
for text, FLOAT/DECIMAL for decimals)
CREATE TABLE book_info (
    BookID INT PRIMARY KEY,
    BookTitle VARCHAR(100),
    Author VARCHAR(100),
    Category VARCHAR(100),
    Price INT,
    Stock INT
);

-- 3. Insert all rows in ONE single command!
-- (Note: If text has an apostrophe like Cox's Bazar, use
double quotes "Cox's Bazar")
INSERT INTO book_info (BookID, BookTitle, Author, Category,
Price, Stock) VALUES
(1, 'Web Basics', 'John Smith', 'Programming', 450, 10),
(2, 'Database Guide', 'Alice Roy', 'Programming', 550, 8),
(3, 'English Grammar', 'Mary Khan', 'Language', 300, 15),
(4, 'Business Math', 'David Lee', 'Business', 400, 12),
(5, 'Python Mastery', 'Sara Ahmed', 'Programming', 600, 5),
(6, 'French Made Easy', 'Paul Costa', 'Language', 350, 7);
```

```
// In XAMPP, the default MySQL settings are ALWAYS:
// Host: "localhost" | User: "root" | Password: "" (empty
string—no password!)
```

```php
<?php
$conn = new mysqli("localhost", "root", "", "uiu_bookshop");

if ($conn->connect_error) {
    die("Connection Failed: " . $conn->connect_error);
}

// Example 1: SELECT with WHERE, AND/OR, ORDER BY (ASC /
DESC)
// Display BookTitle, Author, and Price where Price > 400 AND
Category = 'Programming'
$sql = "SELECT BookTitle, Author, Price FROM book_info WHERE
Price > 400 AND Category = 'Programming' ORDER BY Price
DESC";
$result = $conn->query($sql);

while ($row = $result->fetch_assoc()) {
    echo $row["BookTitle"] . " - " . $row["Author"] . " - " .
$row["Price"] . "<br>";
}

// Example 2: SELECT with GROUP BY, COUNT/SUM/AVG, and HAVING
$sql = "SELECT Category, COUNT(*) AS Total_books, SUM(Price *
Stock) AS Net_worth FROM book_info GROUP BY Category HAVING
COUNT(*) > 1";
$result = $conn->query($sql);

while ($row = $result->fetch_assoc()) {
    echo $row["Category"] . " - " . $row["Total_books"] . " -
" . $row["Net_worth"] . "<br>";
}

// Example 3: UPDATE Query (10% reduction -> * 0.90 | 10%
increase -> * 1.10)
$sql = "UPDATE book_info SET Price = Price * 0.90 WHERE
Category = 'Language'";
$conn->query($sql);

// Extra SQL Filters if needed:
// WHERE BookTitle LIKE '%Beach%'         (contains the word
'Beach')
// WHERE Rating BETWEEN 4.0 AND 4.5       (inclusive range)
// WHERE LetterGrade != 'D'               (not equal)
?>
```